



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Metric in $\mathbb{R}^n$ and ℓ^p Space

Hölder's Inequality, Minkowski's Inequality

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CONDUCTED BY

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Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \emptyset$ and $d : X \times X \rightarrow \mathbb{R}$ satisfies, for all $x, y, z \in X$:

$$(M1) \quad d(x, y) \geq 0;$$

$$(M2) \quad d(x, y) = 0 \iff x = y;$$

$$(M3) \quad d(x, y) = d(y, x);$$

$$(M4) \quad d(x, z) \leq d(x, y) + d(y, z).$$

Here d is called a **metric** on X .

Normed Linear Space

Normed Linear Space

A **normed linear space** is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfies, for all $x, y \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(N1) \quad \|x\| \geq 0;$$

$$(N2) \quad \|x\| = 0 \iff x = 0;$$

$$(N3) \quad \|\alpha x\| = |\alpha| \|x\|;$$

$$(N4) \quad \|x + y\| \leq \|x\| + \|y\|.$$

Here $\|\cdot\|$ is called a **norm** on X .

Metric Induced by a Norm

Let $(X, \|\cdot\|)$ be a normed space. Define $d : X \times X \rightarrow \mathbb{R}$ as 

$$d(x, y) = \|x - y\|.$$

Show that d is a metric on X .

Norm and Metric on Real Line

💡 Euclidean Norm on Real Line

Show that $\|\cdot\| : \mathbb{R} \rightarrow \mathbb{R}$ defined as $\|x\| = |x|$ is a norm on $\mathbb{R}$.



💡 Euclidean Metric on Real Line

Show that $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$d(x, y) = |x - y|$$

is a metric on $\mathbb{R}$.



Norm and Metric on $\mathbb{R}^2$

💡 Standard Euclidean Norm on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking a ordered pairs of real numbers, $x = (x_1, x_2)$ and defining the *Euclidean norm* by

$$\|x\| = \sqrt{x_1^2 + x_2^2}.$$

Show that $\|\cdot\|$ is a norm on $\mathbb{R}^2$.

💡 Standard Euclidean Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and defining the *Euclidean metric* by

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

Show that d is a metric on $\mathbb{R}^2$.

Norm and Metric Space $\mathbb{R}^n$

💡 Standard Euclidean Norm in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking a ordered n-tuples of real numbers,

$$x = (x_1, x_2, \dots, x_n),$$

and defining the *Euclidean norm* by

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

Show that $\|\cdot\|$ is a norm in $\mathbb{R}^n$.

💡 Standard Euclidean Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n -tuples of real numbers,

$$x = (x_1, x_2, \dots, x_n), \quad y = (y_1, y_2, \dots, y_n),$$

and defining the *Euclidean metric* by

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}.$$

Show that d is a metric in $\mathbb{R}^n$.

Sequence Spaces ℓ^p

Sequence Spaces ℓ^p

For $1 \leq p < \infty$, the **sequence space** ℓ^p is defined as

$$\ell^p = \left\{ x = (x_1, x_2, x_3, \dots) : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}.$$

Define

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}}, \quad x \in \ell^p.$$

Show that $(\ell^p, \|\cdot\|_p)$ is a normed linear space.

Hölder's Inequality

💡 Hölder's Inequality

Let $x = \{x_i\} \in \ell^p$ and $y = \{y_i\} \in \ell^q$ be sequences of real numbers, where $p > 1$ and q is the conjugate exponent of p , that is,

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then Hölder's inequality states that

$$\sum_{i=1}^{\infty} |x_i y_i| \leq \left(\sum_{j=1}^{\infty} |x_j|^p \right)^{1/p} \left(\sum_{k=1}^{\infty} |y_k|^q \right)^{1/q}.$$

Minkowski's Inequality

💡 Minkowski's Inequality

Let $x = \{x_i\} \in \ell^p$ and $y = \{y_i\} \in \ell^p$ be two sequences of real numbers, where $p \geq 1$ is a real number. Then Minkowski's inequality states that

$$\left(\sum_{i=1}^{\infty} |x_i + y_i|^p \right)^{1/p} \leq \left(\sum_{j=1}^{\infty} |x_j|^p \right)^{1/p} + \left(\sum_{k=1}^{\infty} |y_k|^p \right)^{1/p}.$$

in other words,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Metric in ℓ^p Space

💡 Metric Induced by ℓ^p Norm

Let $(\ell^p, \|\cdot\|_p)$ be a normed space. Define $d : \ell^p \times \ell^p \rightarrow \mathbb{R}$ as

$$d(x, y) = \|x - y\|_p = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{1/p}.$$

Show that d is a metric on ℓ^p .

The ℓ^∞ Space

ℓ^∞ Space

The **sequence space** ℓ^∞ is defined as

$$\ell^\infty = \left\{ x = (x_1, x_2, x_3, \dots) : \sup_{n \in \mathbb{N}} |x_n| < \infty \right\}.$$

Define

$$\|x\|_\infty = \sup_{n \in \mathbb{N}} |x_n|, \quad x \in \ell^\infty.$$

Show that $(\ell^\infty, \|\cdot\|_\infty)$ is a normed linear space.

💡 Minkowski's Inequality in ℓ^∞ Space

Let $x = \{x_i\} \in \ell^\infty$ and $y = \{y_i\} \in \ell^\infty$ be two sequences of real numbers. Then Minkowski's inequality states that

$$\|x + y\|_\infty \leq \|x\|_\infty + \|y\|_\infty.$$

Metric Induced by ℓ^∞ Norm

💡 Metric Induced by ℓ^∞ Norm

Let $(\ell^\infty, \|\cdot\|_\infty)$ be a normed space. Define $d : \ell^\infty \times \ell^\infty \rightarrow \mathbb{R}$ as 

$$d(x, y) = \|x - y\|_\infty = \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Show that d is a metric on ℓ^∞ .

Metric on $\mathbb{R}^2$

💡 Standard Euclidean Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and defining the *Euclidean metric* by

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

Show that d is a metric on $\mathbb{R}^2$.

💡 Taxicab Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and defining the *Euclidean metric* by

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

Show that d is a metric on $\mathbb{R}^2$.

d_p Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and defining the *Euclidean metric* by

$$d_p(x, y) = (|x_1 - y_1|^p + |x_2 - y_2|^p)^{\frac{1}{p}} \quad (p \geq 1).$$

Show that d_p is a metric on $\mathbb{R}^2$.

d_∞ Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and defining the *Euclidean metric* by

$$d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}.$$

Show that d_∞ is a metric on $\mathbb{R}^2$.

Metric Space $\mathbb{R}^n$

💡 Standard Euclidean Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n-tuples of real numbers,

$$x = (x_1, x_2, \dots, x_n), \quad y = (y_1, y_2, \dots, y_n),$$

and defining the *Euclidean metric* by

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}.$$

Show that d is a metric in $\mathbb{R}^n$.

d_p Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n -tuples of real numbers, 

$$x = (\xi_1, \xi_2, \dots, \xi_n), \quad y = (\eta_1, \eta_2, \dots, \eta_n),$$

and defining the *Euclidean metric* by

$$d_p(x, y) = \left(\sum_{i=1}^n |\xi_i - \eta_i|^p \right)^{\frac{1}{p}} \quad (p \geq 1).$$

Show that d_p is a metric in $\mathbb{R}^n$.

d_∞ Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n-tuples of real numbers,

$$x = (\xi_1, \xi_2, \dots, \xi_n), \quad y = (\eta_1, \eta_2, \dots, \eta_n),$$

and defining the *Euclidean metric* by

$$d_\infty(x, y) = \max\{|\xi_1 - \eta_1|, |\xi_2 - \eta_2|, \dots, |\xi_n - \eta_n|\}.$$

Show that d_∞ is a metric in $\mathbb{R}^n$.

Theorem

Prove that the function $d_\infty : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$d'(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$$

where $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$, is a metric on $\mathbb{R}^n$. Also prove that

$$d_\infty(x, y) \leq d(x, y) \leq \sqrt{n} d_\infty(x, y), \quad \forall x, y \in \mathbb{R}^n,$$

where d denotes the standard Euclidean metric on $\mathbb{R}^n$.

Thank You!

We'd love your questions and feedback.

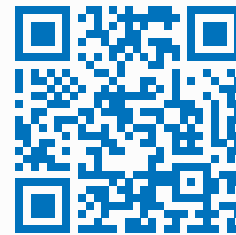
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(Lectures, walkthroughs, and course updates)



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References

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- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.